

Scalability & Future Plan

Date	03 October 2026
Team ID	SWTID-2026-5999
Project Name	PocketSmart AI — AI-Assisted Goal-Based Budget Planner
Maximum Marks	1 Mark

Current Limitations

S.No	Limitation	Effect
1	SQLite is used for the current deployment	Suitable for a small demo but not ideal for many simultaneous writes
2	Shopping destinations are search links	The application does not know live stock or final checkout price
3	Sessions are held in application memory	A multi-instance deployment would require shared session storage
4	Gemini response time depends on the external service	User-facing latency can vary with network and model availability

Scalability Plan

Area	Current Approach	Future Upgrade
Users	Single application instance	Multiple workers behind a reverse proxy
Database	SQLite	PostgreSQL with connection pooling
Sessions	In-memory	Redis or another shared session store
AI requests	Synchronous work handled by the application	Async client, queueing and caching
Files	Local upload directory	Object storage with controlled access
Monitoring	Local test output	Central logging, metrics and error tracking

Future Roadmap

Phase	Enhancement	Expected Benefit
Phase 2	Live retailer/product data	More accurate price comparisons
Phase 3	Cloud deployment and CI/CD	Easier public access and repeatable releases
Phase 4	Editable plans and alternative scenarios	Users can compare different spending strategies
Phase 5	More planning domains such as travel and education	Broader usefulness of the budget-first approach

The roadmap keeps the original idea intact: expand the information available to the planner without removing the backend budget verification step.